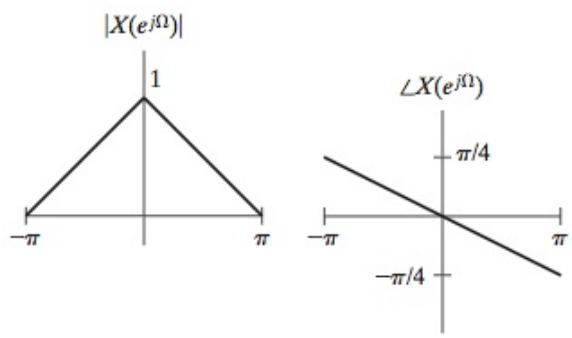
Middle East Technical University
Department of Electrical and Electronics Engineering
EE301: Signals and Systems
Fall 2017-2018

Homework 5
due: 23:55, Jan. 7

Reading assignment: Following sections from Oppenheim: 3.6-3.8, 5.0-5.9, 7.4

1. Let $X(\Omega)$ be the Fourier transform of the DT signal $x[n]$. The magnitude and phase of $X(\Omega)$ are shown below for $|\Omega| < \pi$.



   (a) Give a fully labeled sketch of the magnitude and phase of the DTFT of $x[-n]$ over the range $-\pi < \Omega < \pi$.

   (b) Give a fully labeled sketch of the magnitude and phase of the DTFT of $x[n-1]$ over the range $-\pi < \Omega < \pi$.

2. (2015 Final Exam Question) In this problem, you will explore the aspects of discrete-time filtering. Parts (a) and (b) are independent and may be solved in either order.

   Consider a discrete-time LTI system which takes an input signal $x[n]$, and passes it through a filter with impulse response $h[n]$.

   (a) We want to design a new filter in the following manner. The impulse response values of the new filter, $g[n]$, are related to $h[n]$ as follows:

$$g[n] = \begin{cases} -h[n], & \text{if } n \neq 0 \\ 1 - h[0], & \text{if } n = 0 \end{cases}$$

   Determine a simple expression for $G(\Omega)$, the frequency response of the new filter, in terms of $H(\Omega)$. If the original filter with impulse response $h[n]$ is a low-pass filter with unit gain, i.e. $|H(0) = 1|$, what kind of a filter is this?

   (b) Set $h[n] = \frac{1}{2}\text{sinc}\left(\frac{n}{2}\right)$. Let the input signal be a modulated signal given by $x[n] = z[n]w[n]$ where $z[n] = \frac{1}{2}\text{sinc}\left(\frac{n}{2}\right)$ and $w[n] = \cos\left(\frac{\pi}{2}n\right)$. Plot the spectrum $X(\Omega)$ and determine the output signal $y[n]$. Hint: $\frac{1}{2}\text{sinc}\left(\frac{n}{2}\right) = \frac{\sin(\pi n/2)}{\pi n} \longleftrightarrow \text{rect}\left(\frac{\Omega}{\pi}\right)$

3. (a) Consider the following discrete-time signals:

$$x_1[n] = 2.5cos(n\pi/3 + 2.5)$$

and

$$x_2[n] = 3sin(4n/5 - 1) + 3.$$

Are these signals periodic discrete-time signals? If either of these signals is periodic, determine its fundamental period and DTFS coefficients. Also plot the DTFS coefficients for all k, i.e. from $-\infty$ to ∞.

(b) Let $x[n]$ be a periodic signal with fundamental period N and DTFS coefficients a_k. Express each of the following signals in terms of a_k.

i. $x[n]\cos(\pi n)$
Hint: First find the DTFS coefficients of this signal in terms of a_k.

ii. $y[n] = \begin{cases} 0, & n \text{ even} \\ x[n], & n \text{ odd} \end{cases}$
Hint: Use the result of part (i).

4. Let

$$x[n] = \begin{cases} 1, & n = 0, 1 \\ -1, & n = 14, \\ 0, & 2 \leq n \leq 13 \end{cases}$$

be a periodic signal with fundamental period $N = 15$.

(a) Plot the signal $x[n]$ in the interval $[-20, 20]$.

(b) Find the DTFS coefficients of $x[n]$ analytically.

(c) Plot the DTFS coefficients a_k obtained in part (b) in MATLAB for $-7 \leq k \leq 7$.

(d) Let's define an approximation to $x[n]$ as follows:

$$x_M[n] = \sum_{k=-M}^{M} a_k e^{jk(2\pi/N)n}$$

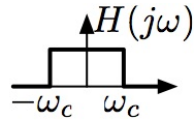
which includes up to and including Mth harmonic components. In MATLAB, plot the approximation $x_M[n]$ using only

i. the DC component (i.e. $M = 0$)
ii. upto and including the fundamental components
iii. upto and including the 2nd harmonic components
iv. upto and including the 4th harmonic components
v. upto and including the 6th harmonic components
vi. including all harmonic components

Comment on your results.

(e) Remember the Parseval's relation for discrete-time periodic signals. First, compute the left part of the relation analytically, which gives the average power of the periodic signal $x[n]$. Next, compute in MATLAB the right part of the relation where the summation in the right part goes from $k = -i$ to $k = i$ for $i = 0, 1, \ldots, 7$ and plot these computed values as a function of i. For which value of i does the computed value exceed 90% of the average power?

5. Consider a continuous-time system with input $x_c(t)$, output $y_c(t)$, and frequency response $H(\omega)$ shown below where $w_c = 2\pi \times 10^3$:



We want to implement this continuous-time filter using discrete-time processing. For this purpose, consider the system shown below which involves discrete-time processing of the continuous-time signal $x_c(t)$ using sampling period T. The C/D operation takes as an input continuous-time signal $x_c(t)$ and gives as output its samples $x[n] = x_c(nT)$. The D/C operation does the opposite, it takes as an input the samples $y[n] = y_c(nT)$ and gives as output the continuous-time signal $y_c(t)$.

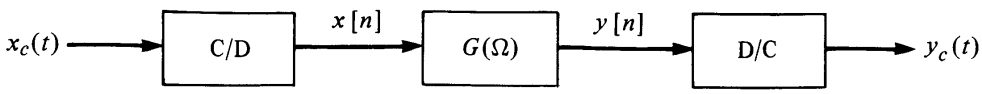
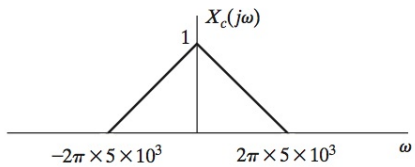


Figure 1: Discrete-time processing of a continuous-time signal

Suppose the spectrum of the continuous-time input signal $x_c(t)$ is as follows:



(a) Find and plot the desired $Y_c(\omega)$ using $H(\omega)$.

(b) Determine the range of sampling frequencies ω_s and sampling periods T that will allow perfect recovery of $x_c(t)$ from its samples $x[n]$.
Assume, for parts (c) and (d), the sampling period T is chosen from this range.

(c) Express the spectrum of $x[n]$ in terms of $X_c(\omega)$ and plot it. Similarly, express the spectrum of $y[n]$ in terms of the desired $Y_c(\omega)$ and plot it.

(d) What is the required discrete-time filter $G(\Omega)$ in Figure 1 to obtain the desired $y_c(t)$ using discrete-time processing? Express $G(\Omega)$ in terms of $H(\omega)$, and plot in the range $[-2\pi, 2\pi]$.

(e) Now suppose a sampling period that does not satisfy the Nyquist sampling theorem was chosen. Comment on what would change in your answers to parts (c) and (d).

6. (Additional question- **WILL NOT BE GRADED**) Compute the DTFT of each of the following signals for $\pi \leq \Omega \leq \pi$

(a) $x[n] = \begin{cases} n, & -3 \leq n \leq 2 \\ 0, & otherwise \end{cases}$

(b) $x[n] = \sin\left(\frac{\pi}{3}n\right) + \cos(n)$

(c) $x[n] = u[n-5] - u[n+1] + 8\delta[n-3]$

(d) $x[n] = \left(\frac{1}{3}\right)^{|n|} u[n+2]$

7. (Additional question- **WILL NOT BE GRADED**) Consider an LTI system with impulse response

$$h[n] = \left(\frac{1}{3}\right)^n \sin\left(\frac{\pi}{6}n\right) u[n]$$

(a) Find the frequency response $H(\Omega)$ of the system and plot in the range $[-2\pi, 2\pi]$.

(b) Suppose that $x[n] = (-1)^n \cos\left(\frac{\pi}{3}n\right)$ is input to this system. Your task in this part is to compute the system output $y[n]$. Observe that there are <u>three</u> possible ways for this task. State and briefly explain all of them. Then find the answer using one of these ways.

(c) Now consider another system whose frequency response is given by $|H(\Omega)|^2$. Express the system's impulse response in terms of $h[n]$.

(d) Repeat part (c) for a system with frequency response $H(\Omega - \Omega_0)$ where Ω_0 is some constant.

8. (Additional question- **WILL NOT BE GRADED**) An LTI system has an impulse response $h[n] = 3^n u[n] + \delta[n+4]$.

(a) Is this system stable? Is it causal?

(b) Does the Fourier transform of $h[n]$ converge? (Equivalently, does the frequency response of this system exist?) Give a justification only using the DTFT definition.

(c) Determine the system (transfer) function $H(z)$ and the associated ROC. Also plot the ROC.

(d) Comment on the stability and causality of the system based on $H(z)$ and its ROC. Also check the consistency of your answers with those in part (a).

(e) Comment on the convergence of the DTFT of $h[n]$ based on $H(z)$ and its ROC. Also check the consistency of your answers with that in part (b).

(f) Let r be a real and nonnegative number, that is $r \geq 0$. State for what range of values of r the Fourier transform of the sequence $r^{-n}h[n]$ converges.

(g) $H(z)$ for $z = 5e^{j(\Omega+\pi)}$ can be viewed as the Fourier transform of a sequence $x[n]$; that is, $H(z)|_{z=5e^{j(\Omega+\pi)}} = \mathcal{F}\{x[n]\}$. First express $x[n]$ in terms of $h[n]$ and then determine its explicit form.

(h) Now consider another LTI system whose impulse response is equal to $(2j)^n h[-n]$. Using z-transform properties, find the system (transfer) function together with its ROC.